



The header features a dark background with two network graphs: a red one on the left and a blue one on the right. In the center, there are six white icons arranged in two rows. The top row includes a head with a brain and pulse line, a server rack, and a scatter plot with a regression line. The bottom row includes a bar chart with an upward arrow, a 2x2 grid of mathematical symbols (+, x, -, ÷), and a diagram of nodes and connections with binary code (1s and 0s) around it.

CIA2C14 Data Science Essentials

Data Modelling: Clustering

Machine learns via two main styles

II. **UN**Supervised Learning:

- Process of an algorithm to model the underlying structure or distribution in the data in order to uncover previously unknown patterns in data
- To bring similar observations together
- There is NO teacher-student relationship
- There is NO feeding of **labels/targets** during training.
- NO way to determine how accurate is the outcome (no Ground Truth to verify)

We are different.
But you need to
study us in detail
to know why



Supervised Vs Unsupervised Learning

	Supervised Learning	Unsupervised Learning
Advantage	Label acts as the ground truth to validate model performance	Unlabelled data are easily available
Disadvantage	Difficult/expensive to obtain labelled data	More subjective than supervised learning, as there can be no simple goal. Won't be able to know if the model is right or wrong, e.g. users may interact more frequently with some particular product listings, BUT that does not mean the user 'like' those products-'like' or 'dislike' is a label.

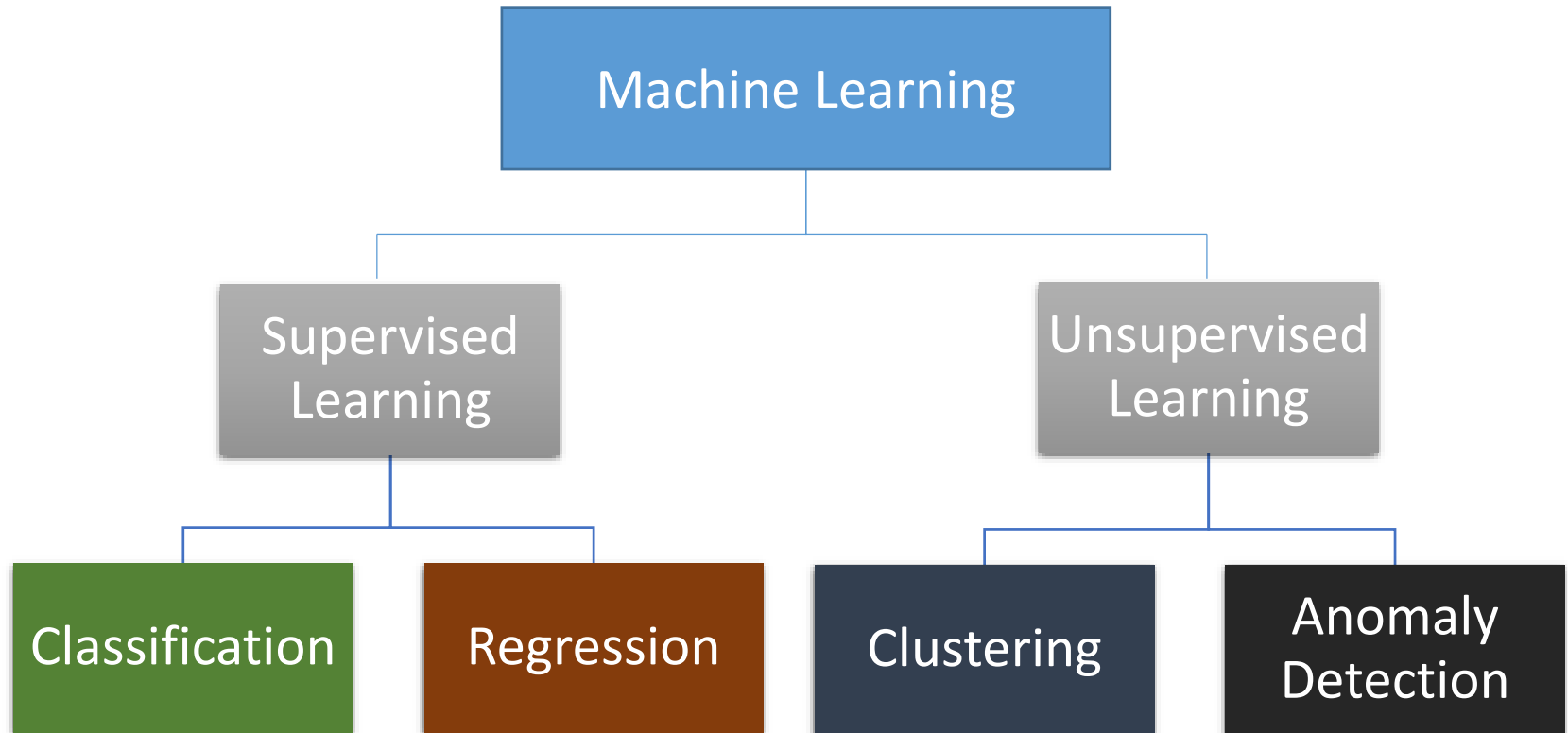
Supervised Vs Unsupervised Learning

- The ability to verify if the model is accurate or not, makes supervised learning applicable to most practical needs in the industry.
- However there are also some applications that Unsupervised Learning can perform better.

Can you name a few?

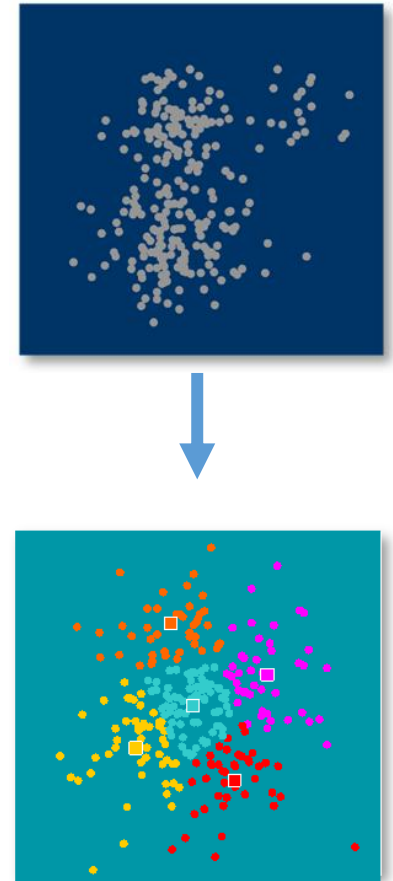


Different families of algorithms

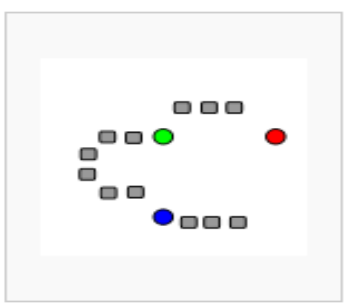


k-means Clustering

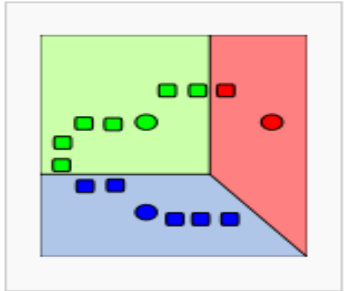
- One of the more commonly used methods for clustering is the **k-means algorithm**.
- Common use of this is to **observe patterns and find clusters/groups**.
- Examples:
 - Market segmentation
 - Profiling claimants
 - Fraud Detection



The Workflow

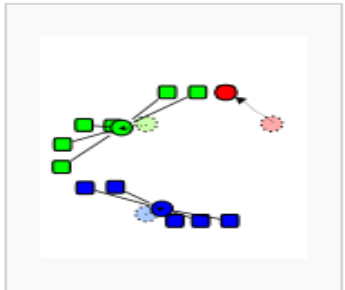


Step 1: Choose k (the number of desired clusters) random positions in the input space

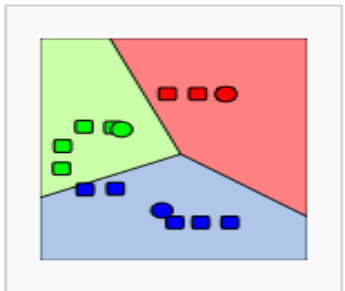


Step 2: Assign them as cluster centroids

Step 3: For each data instance, compute the distance to each centroid



Step 4: Assign the data instance to cluster with min distance

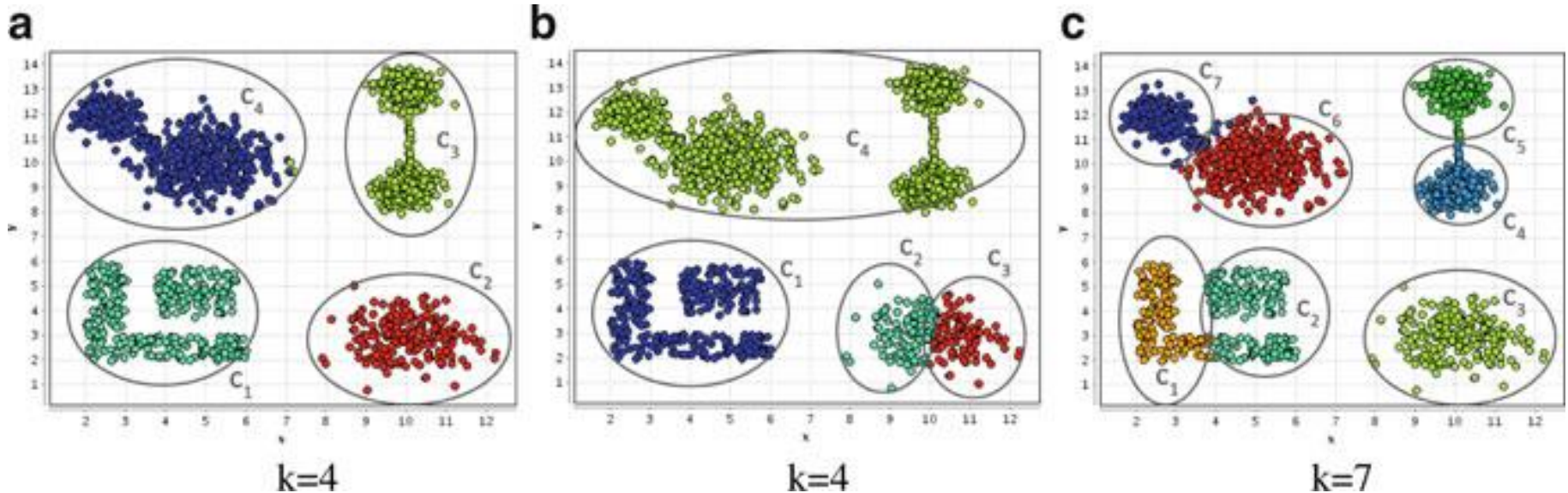


Step 5: Compute mean of each cluster and assign new cluster centroids

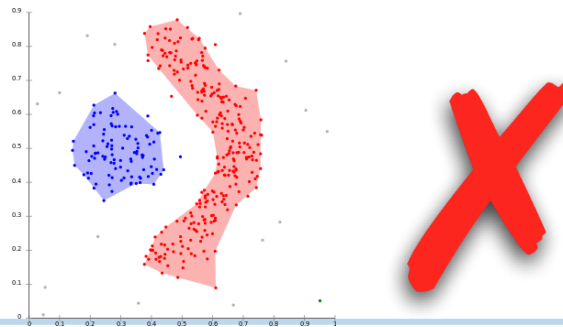
Step 6: Repeat Steps 3-5 until convergence

Challenges of k-means

- What is the suitable k (no. of clusters)?



- Does not work well for overlapping data points



**WE  DATA
SCIENCE**